

A 3-DOF ELECTROMAGNETIC MICRO-ACTUATOR FOR HIGH-RESOLUTION HAPTIC FEEDBACK

ELECTROMECHANICAL DESIGN & ADDITIVE MANUFACTURING

TOOLS & METHODS:

DLP additive manufacturing at 35 μm (Anycubic Photon D2) · custom coil-winding fixture and sacrificial bobbin · Ecoflex elastomer casting · COMSOL Multiphysics (Magnetic Fields, homogenized multi-turn coil) · Arduino PWM with MOSFET gate drive and boost conversion · laser displacement metrology · spring-damper tissue modeling · unit-cost analysis.

EXECUTIVE SUMMARY:

Human touch resolves texture, slip and pressure through a rich mix of frequencies and directions, yet most haptic interfaces still deliver one buzz from an eccentric-mass motor. Closing that gap requires actuators small enough to array at fingertip density, fast enough to span the tactile bandwidth, and able to push sideways (shear) as well as inward (normal).

This programme produced such a node. A DLP-printed cube housing holds a neodymium magnet surrounded by figure-8 copper coils on each face; energizing a coil produces a Lorentz force along that axis, giving three independent degrees of freedom from one part. COMSOL simulations placed the magnet and sized the coils, a custom winding fixture cured a force-linearity problem, and a MOSFET driver chain unlocked >20 V drive from a 5 V microcontroller, at a projected volume cost near \$0.50 per unit.

KEY RESULTS:

- **Three degrees of freedom from one node:** shear in X and Y plus normal Z motion, replacing a four-actuator array and cutting part count to roughly one quarter.
- **Independent axis control** confirmed by laser position tracking as X, Y and both coils were driven in turn.
- **Linear displacement and acceleration response** below ~10 Hz; square-wave drive roughly doubled peak-to-peak displacement relative to sine.
- **Force linearity restored** by a custom wire tensioner and removable DLP-printed bobbin that eliminated non-uniform winding density.
- **> 20 V coil drive** via a logic-level MOSFET stage and boost converter, versus the 5 V ceiling of direct microcontroller drive.
- **~\$0.50 projected unit cost** at volume with outsourced pre-wound coils.

INTRODUCTION:

Vibrotactile feedback is the primary channel through which a wearable or teleoperated system conveys physical contact, but the actuators in general use are eccentric-mass or linear-resonant motors that produce a single, essentially fixed vibration along one axis. Real tactile perception depends on both frequency content and direction: shear motion signals slip and texture, while normal motion signals pressure and contact.

The design objective was therefore a node small enough to tile at fingertip density, capable of independent motion in three axes, and cheap and repeatable enough to manufacture in quantity. An initial cylindrical housing with a single axial coil, tiled into an array with shear coming from coordinated neighbours, proved larger, costlier and harder to control than the problem required.

TEST SETUP:

Architecture. Re-imagining the housing as a cube with a figure-8 coil bonded to each face allows one magnet to be driven along any axis: current in the Z coil lifts it, while the X or Y coils apply a moment that shears it sideways. The change removed three of every four parts while adding two degrees of freedom. Three fabrication constraints shaped the final unit. Irregular winding tension produced a nonlinear force response, corrected with a wire-tensioner fixture and a single-use DLP-printed bobbin broken away after epoxy cure. COMSOL field maps located the magnet where coil flux is strongest without enlarging the housing. An Ecoflex elastomer layer on the contact face improves mechanical coupling to the fingertip and dampens shock against the rigid resin housing.

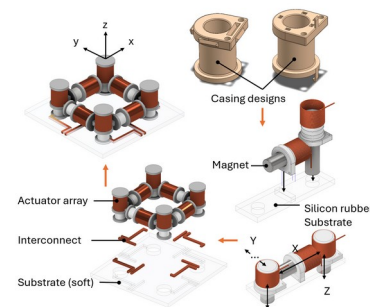


FIG 1: From actuator array to single 3-DoF node: casing iterations, magnet placement, and figure-8 coils on the X, Y and Z faces.

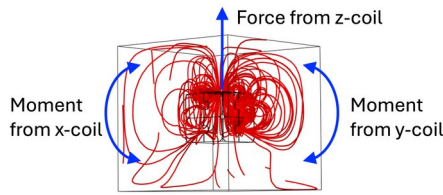


FIG 2: COMSOL magnetic-field streamlines for the cube housing: flux from each face coil and the resulting axial force and shear moments on the central magnet.

Electromagnetic modeling. The housing CAD was imported into COMSOL's Magnetic Fields interface with the windings as a homogenized multi-turn coil of 160 turns of 28 AWG wire, the count taken from the 3.25 mm winding width divided by wire diameter, driven at 1 A inside a 20 cm air domain to suppress boundary effects. Streamline plots confirmed a dipole-like field on the coil axis. Force on the magnet is extracted by the Maxwell stress tensor or by integrating the Lorentz force over the magnet volume, then compared with its ~2 mN weight to set the minimum actuation current. Iterating turn count, wire gauge and coil offset guided the final geometry.

Drive electronics and control. A microcontroller generates PWM whose frequency, amplitude and duty cycle map to the perceived rate, intensity and "press" of each tap, across four waveform families: sinusoidal for smooth textures, square for crisp tapping, triangular for pressure ramps, and an FFT-composite mode summing weighted harmonics to mimic the multi-frequency signature of real surfaces. The first circuit drove coils directly from the microcontroller and was capped at 5 V; the second gates a logic-level MOSFET so a bench supply or boost converter delivers over 20 V across the inductor, giving stronger, faster oscillation. Displacement and acceleration were measured with laser displacement metrology across a frequency sweep.

RESULTS:

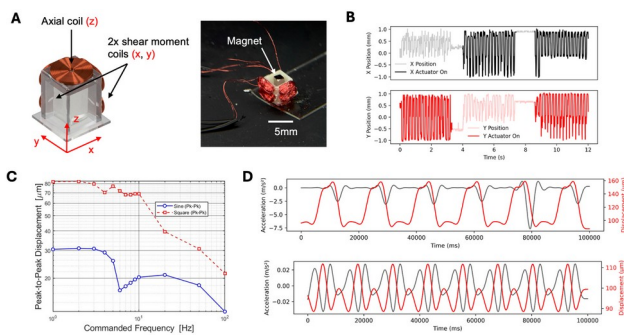


FIG 3: (A) The 3-DoF actuator and fabricated prototype (scale bar 5 mm). (B) X and Y position as each axis is driven in turn. (C) Peak-to-peak displacement versus frequency. (D) Triangular and modulated-sine waveforms.

The prototype meets its objective: a compact, reproducibly manufactured node delivering independent X, Y and Z vibrotactile output across a continuous frequency range. Position traces show clean switching between axes as each coil

is energized in turn, displacement and acceleration scale linearly up to approximately 10 Hz, and square-wave excitation yields markedly larger strokes than sine at the same commanded frequency. Above 10 Hz the response rolls off, though this may reflect the measurement setup rather than the actuator itself; that roll-off and the residual X–Y cross-coupling are the two targets for the next revision. A spring–damper model of the node, its Ecoflex collar and the fingertip, with displacement shared between the Ecoflex ($E \approx 55$ kPa, 2 mm shell) and soft tissue ($E \approx 100$ MPa over 4 mm) in proportion to stiffness, sized the force required for a 1 mm stroke at 10 Hz, with the spread of published skin moduli as the dominant uncertainty.

APPLICATIONS:

Each node is printed, wound and assembled in minutes from resin, a magnet, wire and epoxy, at a projected fifty cents per unit with outsourced pre-wound coils. That makes it a credible building block for wearable arrays that translate sensor data into distributed, texture-rich touch rather than undifferentiated vibration. Target applications include VR and AR gloves and controllers, teleoperated robotics and surgical systems where the operator must feel contact and slip, and tactile displays for people with visual impairment.

The near-term path is defined: tile nodes into a patch, close the loop with tactile sensors, evaluate coil geometries that lower drive current, and miniaturize with off-the-shelf micro-coils.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com